
Auditing Gait Laterality in Skeleton Predictive Representations

Anonymous Author(s)

Affiliation

Address

email

Abstract

Human movement provides a useful test of whether predictive representations preserve the geometry of an articulated body. Left-right reflection supplies a known transformation, while differences between the two sides provide an observable quantity whose sign should change under that transformation. We study this relationship in a skeleton Joint-Embedding Predictive Architecture using pose sequences from gait videos. A source-separated evaluation compares trained and initial encoders, tests reflection behavior, and changes the anatomical and temporal placement of hidden training targets. The available experiments show that predicting a clip’s hidden features more accurately than another clip’s features can coexist with weaker recovery of a signed movement contrast. Motion-sensitive summaries improve access to information already present at initialization, while motion-weighted and connected-region masks provide no demonstrated advantage over their matched random references. We distinguish this evidence from exact symmetry imposed at the output and from synthetic demonstrations of explicit reflection training. The study contributes an evaluation of articulated-body representations for Physical World AI, identifying a gap between feature prediction and the preservation of a task-relevant physical measurement.

1 Motivation and research question

Walking involves coordinated movement of an articulated body with corresponding left and right limbs. This geometry provides both useful regularities and meaningful departures from them. An average description of movement can remain unchanged when anatomical sides are exchanged, whereas a quantity describing which side moves more should reverse sign. A representation intended for physical prediction may need to support both responses.

Gait abnormalities motivate studying that distinction. Post-stroke studies measure spatial and temporal asymmetries, and Parkinson’s research examines unequal arm swing Patterson et al. [1], Lewek et al. [2]. These observations motivate our measurement; they do not validate it as a clinical index. Here, laterality means a signed comparison of estimated left- and right-side movement.

We ask whether masked feature prediction makes that comparison easier to recover from a skeleton encoder on held-out source videos. The initial null is that training provides no improvement over the same encoder at initialization under a fixed downstream evaluation. Separate questions concern whether the output changes sign correctly and whether landmark features follow a specified reflection rule. Answering one does not establish the others.

Our contribution is a controlled evaluation of a small predictive representation model. The completed experiments concern whole-clip masked prediction and frozen-feature readouts. Their relevance to world models is the assessment of a geometric property that could matter to a future dy-

namics model. The study has no demonstrated real-data forecasting, planning, or multimodal-fusion result.

2 Data and the geometric target

The local subset of the Gait Abnormality in Video Dataset [3] contains 666 annotations, of which 642 have pose archives. Quality control retains 625 sequences from 93 source videos. The annotations cover normal, Parkinson’s, stroke, myopathic, and cerebral-palsy gait. They stratify source assignments and describe the dataset; they do not supervise encoder training.

Each archive contains a variable-length sequence of 33 landmarks with estimated coordinates and visibility. Coordinates are derived from video, including an inferred depth component, and are not calibrated metric three-dimensional motion capture.

For each of five bilateral pairs—shoulders, knees, ankles, heels, and foot tips—we calculate left and right speeds using observed timestamps and only transitions for which both landmarks are observed at both ends. Let their median speeds be $m_{L,k}$ and $m_{R,k}$. The target is

$$y(x) = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + \epsilon}.$$

All five pairs must provide at least eight shared valid transitions. Hips establish the pelvis reference but are excluded from this target: pelvis centering makes their speed magnitudes identical. The target is calculated before input interpolation and resizing.

Reflection M negates the centered horizontal coordinate and exchanges all anatomically paired landmarks, together with validity. Consequently, $M^2x = x$ and $y(Mx) = -y(x)$. This identity is a property of the coordinate operation and formula, rather than evidence that every reflected recording would occur naturally.

The encoder input is prepared separately. Visibility at least 0.45 and finite coordinates determine observations; up to four missing samples can be interpolated for input only. Pelvis normalization removes translation and applies a body scale. Resizing then samples 64 locations along normalized sequence index, without preserving the original timestamp intervals. Values under invalid entries are zero-filled only alongside an explicit validity mask.

Stage	Array shape	What remains associated; what changes
Archived clip	$T_i \times 33 \times 4$, plus frame numbers and frame rate	Coordinate channels and visibility remain linked to one source and sequence.
Target branch	$T_i \times 33 \times 3$, validity $T_i \times 33$	Original timestamps and observed-only paired transitions determine the target.
Model input	$64 \times 33 \times 3$, validity 64×33	Joint identities remain; interpolation, normalization and resampling alter coordinate values and timing.
Four-frame patches	$16 \times 33 \times 12$	Each patch concatenates four coordinates for the same landmark; complete validity determines eligibility.
Embedded token grid	$16 \times 33 \times 96$	Time-block and landmark indices identify each 96-dimensional feature.
Batch with mask	$B \times 528 \times 96$, plus target indices and validity	Gathering and padding retain each clip’s own targets and common mask count.

The pipeline preserves the correspondence between recordings, landmarks, masks, and targets. It cannot promise preservation of the exact observed shape or speed after preparation. Cohort preparation occurs before split-file creation in the implementation; it uses deterministic clip-local operations rather than statistics fitted across sources. Source membership is fixed before training sampling, augmentation, and model fitting.

3 Training and source separation

All clips and derived views from a source video inherit its fold. Five outer folds contain respectively 19, 19, 19, 18, and 18 test sources; their test sequence counts are 189, 182, 72, 77, and 105. Each encoder trains only on its matching outer-training sources. New encoders are initialized for each fold and seed, and source-balanced sampling limits domination by videos with many clips.

The base model uses a 96-dimensional embedding, four encoder layers, two predictor layers, and four attention heads. The online encoder receives a full token grid with hidden coordinate content suppressed and positional information retained; a predictor estimates teacher features at hidden positions. A target encoder supplies those features and is updated by an exponential moving average (EMA) of online weights. Its parameters receive no gradients.

The real-data objective matches distributions over feature channels using cross-entropy,

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(p_{\bar{\theta}}, p_{\theta, \phi}) + 0.05 \mathcal{L}_{\text{VICReg}}.$$

Here $\theta, \phi, \bar{\theta}$ denote online-encoder, predictor, and teacher parameters. Teacher centering and separate temperatures form the target and predicted distributions. VICReg penalizes insufficient feature variation and redundant channels, alongside agreement between geometric views. It is calculated from unmasked views pooled over twelve gait landmarks. Consequently, anatomical information remains in the regularizer even when targets are chosen across the body. Mean-squared feature error appears in diagnostics; it is not the real-data pretraining objective.

The historical base profile records 300 epochs, four updates per epoch, AdamW, batch size 20, learning rate 10^{-3} , and weight decay 0.05. All completed controlled grids use 1,200 updates per encoder. Mask fraction is applied to eligible gait tokens and reduced to a feasible shared count; the original 0.6 setting does not hide 60% of all 528 tokens.

The study introduces anatomical laterality at different locations that should be kept explicit:

Entry point	Effect on training or evaluation	Evidence status
Named left/right landmarks and gait pooling	Preserve anatomical identities and choose locations contributing to regularization	Used in completed real-data training
Gait-only target eligibility	Select hidden tokens from six bilateral landmark pairs while retaining all-body context	Historical base and Notebook 12 gait arm
Anatomical reflection augmentation	Exchange joint identities, horizontal signs, validity and transformed views consistently	Previously reported original variant
Explicit reflection penalty	Penalize token disagreement after anatomical exchange; gradients update the online encoder	Notebook 09 synthetic demonstration
Odd readout and signed regression target	Constrain or evaluate a frozen representation	Downstream only

The latest motion/region trainer fixes reflection probability and explicit symmetry weight to zero. It does not test a combined motion-plus-reflection recipe. None of these encoder objectives receives y , a diagnosis, or an affected-side label.

After pretraining, encoder weights are frozen. Inner source-separated validation selects a ridge-regression penalty; ridge regression is a linear readout whose weight penalty limits fitting unstable feature combinations. The original audit uses four inner folds, while later comparisons use three. Outer-training groups contain 74 or 75 sources. Each original inner fit contains 55–57 sources, and validation contains 18 or 19. No outer-test source enters these fits. Imputation and scaling are fitted within each inner fitting partition and refitted on the outer-training set after selection. These inner folds select the readout only: the encoder has already seen their unlabeled inputs during outer-training pretraining.

99 4 Evaluation through successive questions

100 4.1 Does the prepared input preserve the measurement?

101 The original audit prompted a check before attributing weak performance to the encoder. Notebook
102 07 recomputes the movement formula on the prepared input. Repeating that descriptive calculation
103 during this review produces finite pairs for 623 clips from 92 sources. Source-weighted sign agree-
104 ment with the original target is 70.4%; the direct calculation-agreement R^2 is 0.218. Two clips
105 cannot supply a finite recomputation.

106 This comparison changes the entire preparation path, including timing, interpolation and valid sup-
107 port. It identifies a measurement mismatch without localizing its cause. Its R^2 is neither a model
108 score nor an information ceiling: a learned mapping from prepared input could behave differently.
109 It motivates evaluating the target after individual preparation stages on the same eligible clips.

110 4.2 Does the original representation satisfy the reflection audit?

111 The retained original aggregate summary reports source-balanced $R^2 = 0.060$, with a 95% source-
112 bootstrap interval of $[-0.025, 0.126]$. The learned-minus-initial difference is -0.018 , with inter-
113 val $[-0.039, 0.002]$. Reflection augmentation changes R^2 by 0.004, with interval $[-0.006, 0.013]$.
114 These estimates do not demonstrate a training benefit or establish equivalence between variants.

115 The direct token test aligns landmark positions under reflection while leaving feature channels un-
116 changed. Its normalized squared discrepancy is

$$q = \frac{\|Z(Mx) - SZ(x)\|_C^2}{\|Z(Mx)\|_C^2 + \|SZ(x)\|_C^2},$$

117 where S exchanges joint identities and C selects common valid tokens. The retained summary re-
118 ports $q = 0.114$ for learned features and 0.083 at initialization, with a paired increase of 0.031,
119 interval $[0.016, 0.048]$. This is unfavorable evidence for the specified identity-channel transfor-
120 mation. An encoder could express reflection through a different transformation of feature channels; the
121 test does not examine that possibility.

122 An output can be made antisymmetric through an odd feature, $z^-(x) = [z(x) - z(Mx)]/\sqrt{2}$,
123 followed by a linear readout without an intercept or feature centering. Its symmetry follows alge-
124 braically. Predictive accuracy still requires evidence: the recorded learned-minus-initial difference
125 under this construction is negative.

126 The original values are available in a retained figure-summary JSON. Its full report, checkpoint,
127 and per-example prediction chain was not found in this checkout during the present review. We
128 retain these results as previously reported context and distinguish them from the newly audited
129 comparisons below.

130 4.3 Does anatomical target selection improve learning?

131 Notebook 12 compares scattered hidden targets selected from twelve gait landmarks with the same
132 number selected across all 33 landmarks. Both encoders still receive the full landmark set and retain
133 anatomical choices in regularization and readout. Starting weights, sampled clips, geometric views,
134 and 1,200 updates are paired.

135 The teacher-feature readouts obtain $R^2 = -0.023$ with gait targets and -0.010 with all-
136 landmark targets; the initial encoder obtains 0.048. The broader-minus-gait difference has inter-
137 val $[-0.018, 0.050]$, leaving both improvement and harm compatible with these recordings. Larger
138 ridge penalties frequently reach the edge of the search range, limiting conclusions about the best
139 achievable readout. The later grid extends the candidates to 10,000, which is still selected by 39%
140 of teacher motion-sensitive fits and 77% of the corresponding online fits. The reported deficits are
141 results of these readout procedures, rather than a bound on what any decoder could recover.

142 Notebook 08's earlier values are retained in the tutorial but excluded from the main comparison:
143 implementation and readout choices changed together before Notebook 12. Likewise, Notebook

12’s direct-pose $R^2 = 0.130$ and the latest 0.0345 come from different evaluation recipes. Their change cannot be attributed to a mask or a single regularization choice.

4.4 Does targeting motion or connected anatomy help?

Notebooks 15–18 extend the comparison to motion-weighted and connected-region targets. The completed grid contains 125 trained encoders, each with 1,200 updates. Motion comparisons and region comparisons have their own random references, matched to the actual hidden count. Equal nominal mask percentages would not ensure equal tasks because missing observations and eligible landmark pools differ.

Frozen features or control	Mean $R^2 \pm$ seed SD	Mean absolute error
Training-source target mean	-0.0106 ± 0.0000	0.046172
Direct prepared-pose summary	0.0345 ± 0.0000	0.044960
Initial encoder, mean summary	0.0708 ± 0.0186	0.044344
Initial encoder, motion-sensitive summary	0.2225 ± 0.0268	0.041546
Motion-uniform teacher, motion-sensitive	0.1137 ± 0.0110	0.043658
MAMP-style teacher, motion-sensitive	0.1129 ± 0.0306	0.043751
Robust-motion teacher, motion-sensitive	0.1142 ± 0.0210	0.043595
Region-uniform teacher, motion-sensitive	0.1094 ± 0.0088	0.044037
Connected-region teacher, motion-sensitive	0.1008 ± 0.0092	0.043756

Each row covers 625 clips and 93 sources. SD describes the five optimization seeds, not sampling uncertainty. The initial encoder, direct-pose, and training-mean controls are reused across arms; their repeated appearance in machine-readable tables is not independent replication. “Initial” refers only to the S-JEPA weights: the pipeline still uses a pretrained pose detector, supplied anatomical identities, feature engineering, and a supervised ridge readout.

The motion-sensitive summary adds temporal variation, feature changes, and observation support. All trained arms remain below their matched initial encoder in each of the five seeds. Its advantage over mean pooling cannot yet be attributed solely to motion because the summary also includes support information. Temporal standard deviation is insensitive to ordering. The initial gain is 0.1517 in R^2 , compared with 0.0270–0.0468 for trained teachers. The online encoders also underperform initialization, with mean R^2 between 0.0636 and 0.0809, so evaluating the teacher alone does not explain the deficit.

We additionally recomputed trained-minus-initial intervals from the saved predictions for this revision. They are new exploratory analyses, using 2,000 paired source resamples with bootstrap seed 812 and no model retraining. All five intervals lie below zero; the motion-uniform contrast is -0.1088 , interval $[-0.1701, -0.0412]$. The full set appears in Appendix A. This strengthens the conditional description of a learning deficit while leaving evaluation choices and adaptive development as limitations.

Mask minus its matched random reference	ΔR^2	95% source interval
MAMP-style motion weighting	-0.0008	$[-0.0204, 0.0153]$
Robust motion mixture	0.0005	$[-0.0155, 0.0152]$
Connected region	-0.0087	$[-0.0333, 0.0177]$

The mask audits confirm that the proposed changes took effect. Motion targets average 80.771 valid tokens per clip, approximately 17% of valid all-body tokens. Relative to eligible-token motion, average enrichment is approximately zero for random selection, 0.0175 for MAMP-style weighting, and 0.0378 for the robust mixture. The mixture retains a uniform component and limits extreme motion scores; high displacement may still reflect tracking error.

In the structure audit, immediate visible time brackets surround 69.9% of random-reference targets and none of the connected-region targets. Connected regions also reduce the availability of visible anatomical neighbors. Thus the masks remove different observable clues, although they do not isolate what the contextualized teacher encodes. Whole trajectories and interior gaps are audited for coverage, but are absent from the completed training comparison.

180 The motion and region families use different hidden budgets and separate random references. We
 181 compare each alternative only with its own reference. The largest numerical score among arms is
 182 not a validated best mask, and an interval spanning zero does not establish equal performance.

183 4.5 What does successful feature prediction establish?

184 The trained predictors consistently predict their own clip’s hidden teacher features more accurately
 185 than targets from another source video, comparing matched and mismatched errors on the same
 186 eligible control clips and target positions. Initial predictors show no consistent preference. This
 187 establishes useful clip correspondence for that diagnostic. Posture, viewpoint, observation support,
 188 or motion could contribute to it; the result does not isolate a physical mechanism.

189 Thus, predictor training can succeed on its own features while the frozen representation yields
 190 weaker laterality readouts. Because each arm supplies its own changing teacher, raw feature-
 191 prediction losses are not a common between-model measurement scale. The coordinate-derived
 192 outcome supplies a shared evaluation target.

193 5 Statistical interpretation

194 For V sources and n_v clips from source v , each clip has weight $w_{vi} = 1/(Vn_v)$. With $\bar{y}_w =$
 195 $\sum w_{vi}y_{vi}$,

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}.$$

196 Each source therefore receives equal total weight even when it supplies many clips. Within a seed,
 197 predictions from all five held-out folds are pooled before computing R^2 ; the reported result averages
 198 the five seed-specific scores. It is neither an average of fold R^2 values nor a score from an ensemble
 199 of seed-averaged predictions. Negative R^2 means squared error exceeds variation around the
 200 weighted evaluation mean; a separately fitted training-source mean is the usable prediction baseline.

201 Paired uncertainty resamples entire videos while retaining their clips and paired seed predictions.
 202 For a mask comparison the null is no positive difference in source-balanced readout performance rel-
 203 ative to the count-matched random policy; for learning it is no positive difference relative to matched
 204 initialization. These are superiority questions, and no equivalence margin or power guarantee is es-
 205 tablished. The 2,000-resample intervals condition on the fitted models and exclude full-retraining
 206 and new-split uncertainty. Repeated seeds and diagnostic mask draws are not independent partic-
 207 ipants. Subsequent experiments were informed by the same development cohort, so their source
 208 separation does not make the overall research trajectory an untouched confirmatory study.

209 6 Implications and limitations

210 The strongest supported conclusion concerns access to one signed movement measurement under
 211 the evaluated readouts. A negative learned-over-initial contrast does not demonstrate that all mo-
 212 tion information has disappeared. Mean pooling, feature scaling, regularization, and the difference
 213 between the original target and prepared input remain possible contributors.

214 The study fits the workshop’s articulated geometry and evaluation themes. Coordinates derived
 215 from RGB do not constitute multimodal sensing, and an inferred depth coordinate does not estab-
 216 lish calibrated 3D reconstruction. Clinical interpretation would require independent measurements,
 217 affected-side labels, and participant-separated validation. The source-video split cannot exclude the
 218 same unidentified person appearing in different videos.

219 The next experiments should first isolate support features, broaden training-only ridge selection,
 220 and measure the target after each preparation stage. A later study can test whether explicit geo-
 221 metric training improves common observable future endpoints, using past-only preprocessing and
 222 persistence and velocity baselines. Existing symmetry-training and future-decoding demonstrations
 223 are synthetic; they are not evidence of forecasting gait recordings.

224 Institutional ethics, data-use, and derived-pose release reviews remain unresolved in the recorded
 225 governance status. This manuscript is an internal revision, and it does not claim those determinations
 226 have been obtained.

227 **7 Related work and reproducibility**

228 S-JEPA [4] already studies masked skeleton-feature prediction, and MAMP [5] motivates motion-
229 weighted masking. SLiM [6] provides recent structured skeleton-masking precedent. Our contri-
230 bution is the controlled laterality evaluation, with no claim to invent skeleton JEPA or anatomical
231 masking.

232 The tutorial, source-split reference, core audit, and extension audit document the evidence and its
233 limits. Original notebooks and experiment artifacts are preserved. This revision adds no training run
234 or clinical finding.

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253 **A Additional exploratory learning contrasts**

254 These contrasts were calculated during the 9 September revision from the complete latest prediction
 255 grid. They were not part of the original registered reflection audit. Each compares the final teacher’s
 256 motion-sensitive summary with its matched initial encoder using the same readout family, clips and
 257 seed. Intervals are marginal percentile source-bootstrap intervals conditional on the fitted models;
 258 there is no family-wise multiplicity adjustment.

Training arm	Learned minus initial R^2	95% source interval
Motion uniform	−0.1088	[−0.1701, −0.0412]
MAMP-style motion	−0.1097	[−0.1651, −0.0514]
Robust motion	−0.1083	[−0.1706, −0.0423]
Region uniform	−0.1131	[−0.1662, −0.0596]
Connected region	−0.1218	[−0.1828, −0.0553]

259 The recomputation script checks the prediction coverage and reproduces the recorded pooled scores
 260 before calculating these contrasts. The extension evidence audit records the retained artifact paths
 261 and full-precision values.